Foundations of Computer Programming: Computational Thinking

Learning objectives:

Part 1: The top-down and bottom-up approach

- Understand the top-down and bottom-up approaches to problem-solving.
- Understand the advantages and disadvantages of top-down and bottom-up approaches.
- Recognise the computational thinking concepts and how to apply them.

Part 2: Pseudocode

- Understand what pseudocode is and why it's important.
- Learn the fundamental structure of pseudocode
- Understand how to represent decision-making and repetition in pseudocode.
- Ensure clarity and effectiveness in writing pseudocode
- Identify and correct common errors in pseudocode writing
- Discussion and clarify doubts

Lesson exercise: Week 2

Part 1: The top-down and bottom-up approach.

1. You are organising a university charity event, present your steps in a hierarchical structure using the top-down and bottom-up approaches.

2. In your own words, explain the benefits and drawbacks of the top-down and bottom-up approaches.

Part 2: Pseudocode

1. What is pseudocode?
2. Why Pseudocode is important?
3. Write pseudo code that reads two numbers and multiplies them together and print out their product.
4. Write pseudocode that tells a user that the number they entered is not a 4 or a 5.
5. Prompt the user to enter a number. If the number is between 0 and 10, output "blue". If the number is between 10 and 20, output "black". If the number is between 20 and 30, output "green". For any other number, indicate that it is not a valid colour option.